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Data Article

Microscopic image dataset of plant-parasitic nematode

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ABSTRACT

Plant-parasitic nematodes (PPNs) are a significant threat to agriculture, causing substantial crop losses worldwide. Accurate identification of PPNs is crucial for effective management strategies; however, conventional methods are time-consuming and require extensive expertise. To address this challenge, we present a comprehensive microscopic image dataset of PPNs, focusing on common plant-parasitic genera. The dataset consists of 1,016 high-resolution images categorized into 11 classes, representing a diverse range of nematode morphologies, including Hemicycliophora, Radopholus, Helicotylenchus, Trichodorus, Xiphinema, Criconemoides, Criconema, Pratylenchus, Hirschmanniella, Hoplolaimus, and Meloidogyne. Nematodes were extracted from samples of various crops exhibiting infestation symptoms using a modified Whitehead-Tray method. The extracted nematodes were fixed, cleared, stained, and mounted for microscopic observation. Images were captured using optical cameras attached to a light microscope. This dataset provides a valuable resource for developing advanced AI-based tools to automate nematode identification and support precision agriculture efforts aimed at reducing crop losses caused by PPNs. Despite limitations such as imbalanced class distribution and geographic specificity, this dataset offers a foundation for further research and the development of efficient nematode identification methods.

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Specifications Table

Subject	Biology, Computer Sciences
Specific subject area	Computer vision, Image classification, Deep Learning, Machine Learning, Nematode, Precision Agriculture
Type of data	Image
Data collection	Dataset were obtained from 2020 to 2023. Root samples from rice, banana, tomato, melon, chili, and eggplant plants showing signs of nematode infestation, along with surrounding soil, were collected from agricultural fields in Yogyakarta and Central Java, Indonesia. Nematodes were extracted using a modified Whitehead-Tray method, then observed under a light microscope. Image acquisition for further analysis was conducted at the Nematology Laboratory, Universitas Gadjah Mada.. Two nematologists were then conducted a thorough evaluation of each image and categorized them into eleven distinct classes, including Hemicycliophora, Radopholus, Helicotylenchus, Trichodorus, Xiphinema, Criconemoides, Criconema, Pratylenchus, Hirschmanniella, Hoplolaimus and Meloidogyne. To standardize the dataset for consistency, a preprocessing technique was applied to convert all images to JPG format with a uniform resolution of 1280 × 1024 pixels.
Data source location	The dataset was obtained in Central Java dan Yogyakarta, Indonesia, with the detail location as follows. • Kejajar, Wonosobo, Central Java (Latitude: -7.2092, Longitude: 109.9244) • Berbah, Sleman, Yogyakarta (Latitude: -7.8054, Longitude: 110.4428) • Bugel, Panjatan, Kulonprogo, Yogyakarta (Latitude: -7.9257, Longitude: 110.1593) • Ngipiksari, Pakem, Sleman, Yogyakarta (Latitude: -7.6065, Longitude: 110.4240)
Data accessibility	The datasets are publicly and freely available on Mendeley data repository. Data identification number: 10.17632/cck8yxj3xw.2 Direct URL to data: https://data.mendeley.com/datasets/cck8yxj3xw/2 Code repository for dataset utilization, preprocessing, and machine learning examples is provided at: https://github.com/nhs-research/parasitic-nematode-EfficientNetV2B0
Related research article	N.H. Shabrina, R.A. Lika, S. Indarti, Deep learning models for automatic identification of plant-parasitic nematode, Artif. Intell. Agric. 7 (2023) 1–12. https://doi.org/10.1016/j.aiia.2022.12.002 [1].

1. Value of the Data

- This dataset contains diverse and high-resolution images of plant-parasitic nematodes collected from agricultural fields in Indonesia, capturing key morphological features across 11 genera.
- The dataset supports the development of AI-powered nematode identification tools by providing training data for supervised learning models, such as CNNs. It can be used in building automated diagnostic applications for integration into pest monitoring systems.
- This dataset can serve as a benchmark for training and evaluating machine learning models in a real-world agricultural context by providing an annotated image repository that captures diverse nematode morphologies.
- The dataset has potential relevance in Integrated Pest Management (IPM), where early detection of plant-parasitic nematodes can inform region-specific control strategies and improve crop protection planning.

- Beyond nematology, the dataset may benefit agronomists, particularly in understanding the threat posed by plant-parasitic nematodes to crop cultivation, especially in relation to specific host commodities that are susceptible to nematode infestation.
- The dataset may support bioinformatics research, complementing genomic studies through phenotype-based analysis, and contribute to digital agriculture.
- This dataset can be used for educational and training purposes, such as training early career nematologists, assisting curriculum development, or integrating into existing nematode diagnostic platforms.
- Compared with other publicly available nematode datasets, this collection is unique because of its geographic focus on tropical plant-parasitic nematodes and its annotated diversity of nematode genera, which are often underrepresented in global databases.

2. Background

Plant-parasitic nematodes (PPNs) are microscopic roundworms that live in soil and plant tissues, where they feed on roots and disrupt the ability of plants to absorb water and nutrients. This often leads to stunted growth, yellowing, wilting, and reduced crop yield, making plants more vulnerable to secondary infection. In severe cases, infestations can cause yield losses of up to 80% [2,3], thereby posing a serious threat to agricultural production. However, identifying PPNs is particularly challenging because of their microscopic size, and symptoms are often nonspecific and easily confused with nutrient deficiencies or other plant diseases. Because different nematode species vary in their host range, aggressiveness, and response to treatment, accurate identification is essential for selecting effective management strategies for nematodes. Without proper identification, farmers may struggle to control the infestations, leading to continued crop loss. Developing reliable diagnostic methods is key to mitigating their impact and ensuring sustainable agricultural productivity.

Recent advancements in image processing and computer vision have shown great potential for automating nematode detection and classification. Training machine learning models on reliable datasets has significantly improved identification accuracy [1,4–7]. This technology addresses the challenges of traditional identification methods, which are time-consuming and require extensive expertise and practical experience. By automating and streamlining the identification process, this approach enhances accessibility, efficiency, and accuracy, making nematode diagnostics more feasible for non-experts.

Given the limited availability of datasets dedicated solely to plant-parasitic nematodes (PPNs), we developed a specialized image repository focusing on common plant-parasitic genera to facilitate rapid image-based identification. Unlike broader datasets [8] that include both free-living and parasitic nematodes without clear differentiation, our collection exclusively features only parasitic species. This targeted approach reduces irrelevant data and offers a more complete representation of parasitic genera, thereby allowing more detailed taxonomic and ecological studies. By focusing on parasitic nematodes, our dataset provides a practical and valuable resource for parasitologists and supports the development of efficient identification tools to reduce crop losses caused by PPNs.

This dataset will contribute to addressing the challenges of identifying PPN by providing a specialized, publicly available resource focused exclusively on plant-parasitic genera. With curated annotations by expert nematologists, it offers a foundation for developing advanced AI tools that can automate nematode identification. By bridging the gap between traditional manual methods and modern technology, this dataset supports precision agriculture efforts aimed at reducing losses caused by nematode infestations.

This dataset supports our previously published research on deep learning models for nematode identification [1], which focused on model development and performance evaluation. The current article is dedicated exclusively to describing the dataset in detail, including its organization, data structure, and collection procedures, to promote transparency, reproducibility, and broader reuse by the research community.

Table 1
Class distribution on the dataset.

Genus	No. of samples
Genus Xiphinema	85
Genus Meloidogyne	211
Genus Trichodorus	44
Genus Hirschmanniella	130
Genus Hoplolaimus	151
Genus Criconemoides	103
Genus Hemicycliophora	6
Genus Radopholus	31
Genus Pratylenchus	116
Genus Criconema	4
Genus Helicotylenchus	135
Total	**1016**

3. Data Description

The dataset consists of 1,016 images of plant-parasitic nematodes, categorized into 11 classes with diverse genus distributions, as detailed in Table 1. To ensure consistency, all images were stored in JPG format with a standardized resolution of 1280 × 1024 pixels. The dataset was organized into 11 folders, each corresponding to a specific genus, allowing for structured access and efficient organization. Each folder contains labelled images that facilitate easy access and organization for training machine learning models.

As shown in Table 1, the dataset exhibits an imbalanced class distribution, with 36% of the classes containing fewer than 65 images. This variation is primarily due to the limited distribution of certain nematode species and their dependence on specific host plants for survival and reproduction. Although some classes contain fewer than ten images, the dataset's overall diversity provides a strong foundation for training models to identify various PPNs. To address the scarcity of data in certain classes, data augmentation techniques, such as adjustments in lighting, rotation, and scaling, can be applied to artificially expand the dataset [1,7]. These preprocessing methods are recommended to enhance model performance in real-world applications where nematode morphology and imaging conditions may vary significantly. Moreover, the dataset currently includes only samples collected in Indonesia, which may not adequately capture the global diversity of plant-parasitic nematodes. Broadening the dataset to include samples from various geographical regions would significantly increase its representativeness and overall value. In addition to the class distribution table, we included a bar chart (Fig. 1) to provide a visual representation of the number of images per nematode class.

The morphology of nematodes varies significantly across genera. *Meloidogyne* species exhibit distinct life stages, including slender, vermiform second-stage juveniles (J2), globose females with subterminal vulvae, and elongated males that twist when heat-relaxed, highlighting strong sexual dimorphism. *Hirschmanniella* nematodes are medium to long (1.1–4 mm) and usually straight or ventrally curved after heat, with no clear sexual dimorphism. They feature a long stylet and elongated conoid tails with terminal mucrons. *Pratylenchus* are small (<1 mm), ventrally curved when heated, and have a dark, flattened labial region with a moderately developed stylet and similar appearance in both sexes. *Criconema* species possess robust, annulated bodies and smooth annuli in females, along with short but strong stylets. *Criconemoides* are characterized by a round anterior, conical or rounded tail, retrorse annules, and a forward-oriented anchor-shaped stylet.

Xiphinema nematodes are thin and long (1.3–5 mm), with extremely long stylets and one or two genital tracts; their tails range from filiform to rounded. *Trichodorus* nematodes are cigar-shaped with smooth cuticles, curved onchiostyles, and well-developed reproductive structures, including a mid-body vulva in females. *Helicotylenchus* are small to medium-sized (0.4–1.2 mm), often spiral-shaped post-heat, with posterior vulva and spherical, slightly truncated

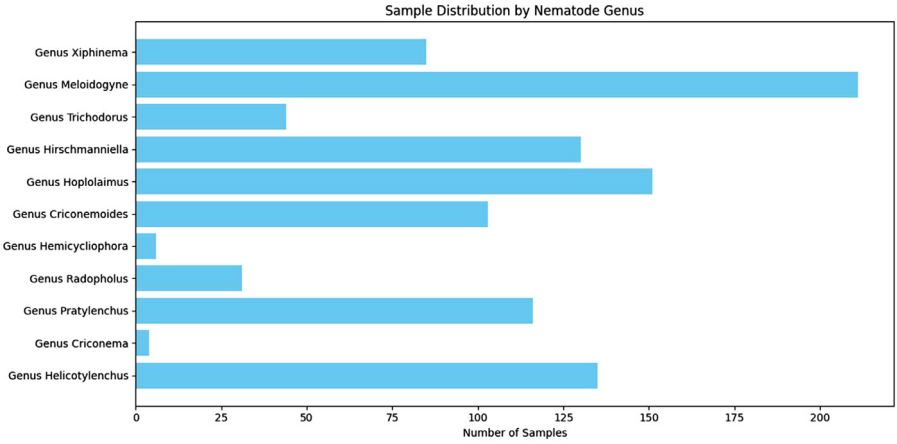


Fig. 1. Distribution of images across the classes.

heads. *Radopholus* are tiny (<1 mm), ventrally curved post-heat, and show strong sexual dimorphism, with males having a taller labial region and reduced feeding structures, whereas females have a conoid tail and a developed reproductive system. *Hoplolaimus* are relatively large (1–2 mm), with offset, heavily sclerotized labial regions, anteriorly directed stylet knobs, and distinct scutella near the vulva. Lastly, *Hemicycliophora* species have a detachable, ornamented cuticle, show sexual dimorphism, and females possess long stylets, pointed tails, and vestigial anus and rectum, with their vulva located near the posterior end.

Table 2 provides a detailed overview of each nematode class, presenting representative sample images alongside descriptions of key morphological features, behavioral traits, and other distinguishing characteristics that determine the placement of each nematode within its respective class [9].

3.1. Data organization

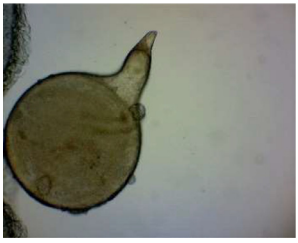
The dataset was organized using a folder-based structure, with each folder corresponding to a specific nematode genus. The images within each folder represent microscopic captures of nematodes identified and labelled by expert nematologists (Fig. 2). This hierarchical structure allows for straightforward navigation, particularly in classification tasks. Each image was stored in JPG format using a consistent naming convention: *genus (image number). jpg. For example, Meloidogyne (1).jpg.*

Accompanying the image data, a metadata file in CSV format is provided, containing general information, such as the file name, label (genus), image size, and format. Although sample-level metadata, such as crop type, sampling date, and geographic location, are available, they cannot be directly linked to individual images. This is because multiple nematode genera were often found within a single sample, and metadata are recorded at the sample level rather than the image level during the data collection process. Future data collection efforts will incorporate more specific metadata to support image-level traceability and enhance contextual analysis.

3.2. Potential ethical considerations

The development and application of AI-based tools for pest management, including nematode identification, must be accompanied by ethical considerations to ensure their responsible and

Table 2
Characteristic and sample dataset.

Genus	Characteristic	Sample images
Genus Meloidogyne	The Meloidogyne genus in this dataset exhibits three distinct phases: second-stage larvae (J2), female adults, and male adults. Both J2 larvae and male adults have vermiform (elongated, cylindrical) bodies of varying lengths, demonstrating sexual dimorphism within the genus. • Juveniles (J2): Vermiform, slender, and approximately 450 µm in length. The stylet and labial region are slightly sclerotized. • Female adults: Globose in shape with a thin neck, measuring between 0.3 to 0.7 mm in diameter. The vulva is subterminal, located near the anus. The stylet is short and moderately sclerotized, while the labial region is weakly structured. • Male adults: Vermiform, measuring 1–2 mm in length, and free-living in soil. Under heat relaxation, the body often twists 180 degrees along its length. The labial region and stylet are robust, with a short, hemispherical tail. Unlike females, males lack a bursa but possess strong spicules.	
Genus Hirschmanniella	Members of the Hirschmaniella genus vary in size from medium to long, ranging from 1.1 to 4 mm. When exposed to gentle heat, they typically die in a straight or ventrally curved position. The anterior region shows no noticeable sexual dimorphism. The labial region is hemispherical or slightly flattened at the front, with annulations that blend seamlessly with the body shape. They possess a well-developed stylet (15–46 µm) with rounded knobs at the base. The elongated esophageal glands extend into a long ventral lobe that overlaps the intestinal tract. In females, the reproductive system consists of two equally developed and functional genital tracts, one directed anteriorly and the other posteriorly. The vulva is positioned along the midline, and the tail is generally elongated, conoid in shape, and ends with a terminal mucron. Males share similar tail characteristics with females, featuring slender, arcuate spicules. However, unlike females, males have a bursa that does not extend to the tip of the tail.	

(*continued on next page*)

Table 2 (*continued*)

Genus	Characteristic	Sample images
Genus Pratylenchus	Pratylenchus is a genus of tiny nematodes, typically less than 1 mm in length. When exposed to mild heat, they die in a slightly ventrally curved position. The anterior region shows no noticeable sexual dimorphism. The labial region is low, flattened, and heavily sclerotized. Under a stereomicroscope, it appears as a dark, flat cap that seamlessly blends with the body contour and is divided into two, three, or four annules. The stylet is moderately sclerotized, measuring no more than 20 µm in length, with rounded or anteriorly concave knobs. Both male and female Pratylenchus nematodes have equally developed esophagi, with a well-formed median bulb.	
Genus Criconema	Several key characteristics define members of the Criconema genus. They have robust bodies with heavily annulated cuticles. Female Criconema lack appendages and have smooth annuli along their bodies. These nematodes possess a sturdy stylet, measuring between 16.5 and 18.5 μm in length, with rounded knobs that slope anteriorly.	
Genus Criconemoides	Criconemoides have a rounded anterior end and a posterior region that ranges from rounded to conical. Their bodies remain straight or slightly curved when relaxed or after death. The cuticle features 42–200 distinct, retrorse annules, with a posterior border that is either smooth or finely crenate. The labial region has one or two thinner annules that blend seamlessly with the rest of the body, lacking clear separation. They possess a robust stylet with basal knobs that form an anchor-shaped structure-oriented forward. The procorpus is fused with a well-developed median bulb in the esophagus, while the glands form a small posterior bulb.	

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Table 2 (*continued*)

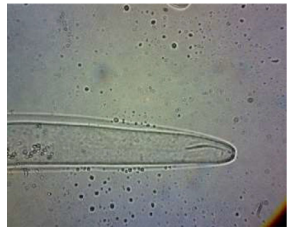
Genus	Characteristic	Sample images
Genus Xiphinema	Genus Xiphinema has thin and elongated bodies ranging from 1.3 to 5 mm in length. The labial region may be either offset or continuous with the body. Their stylet is exceptionally long (60–250 µm) and consists of a needle-like odontostyle with a forked base, which connects to an odontophore featuring three distinct basal flanges. The esophagus comprises a short glandular bulb and a long, narrow procorpus. In females, the vulva is typically located between 40% and 50% of the body length, though in some cases, it is positioned more anteriorly. In individuals with a non-functional anterior reproductive tract (monoopisthodelphic or pseudo-mono-opisthodelphic), the vulva is situated further forward (V = 25%), with the anterior tract being reduced or completely absent. Normally, two fully developed genital tracts are present. The tail shape varies, ranging from long and filiform to short and rounded.	
Genus Trichodorus	Genus Trichodorus is characterized by a robust, cigar-shaped body measuring 0.8–1.2 mm in length, with a smooth cuticle. The labial region is continuous with the body shape, and the papillae are prominent. Their stylet, known as an onchiostyle, is curved and tripartite. The esophagus consists of a thin anterior section and a posterior bulboid expansion. In females, the vulva is located at the mid-body and features strong vaginal sclerotization. The well-developed vagina extends inward to about half the body's diameter. A pair of lateral body pores are positioned within one body diameter of the vulva. Females have two fully developed genital tracts. The tail is short and rounded, with an almost terminal anus. Males possess a gubernaculum and arcuate spicules. The protractor muscles are well-defined, uniquely shaped, and enclose the spicule shafts. The bursa is either absent or extremely small, though ventral supplements are present.	

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Table 2 (*continued*)

Genus	Characteristic	Sample images
Genus Helicotylenchus	Members of the genus Helicotylenchus have small to medium-sized bodies, ranging from 0.4 to 1.2 mm in length. When exposed to heat relaxation, they often die in a spiral shape, though some may adopt a C-shaped posture. The labial region is spherical and conoid, with mild sclerotization and occasional truncation. Their well-developed stylet is typically three to four times the diameter of the lip region, with knobs that are either spherical or cup-shaped. The esophageal gland lobe primarily overlaps the ventral side of the intestine. In females, the posterior reproductive branch is rarely reduced or non-functional. The vulva is positioned toward the posterior end (60–70% of body length), and both genital tracts are typically fully developed [10].	
Genus Radopholus	Genus Radopholus is tiny nematode, typically less than 1 mm in length. When exposed to heat relaxation, their bodies become slightly curved or bent ventrally rather than remaining straight. Sexual dimorphism is prominent in the anterior region. In males, the labial region is taller, often knob-like, and more offset from the body contour. In contrast, females have a lower, rounded labial region that is either continuous with the body or only slightly offset. Females exhibit strong cephalic sclerotization, along with a well-developed stylet and esophagus. Males, however, have pronounced labial sclerotization but a reduced stylet and esophagus. The esophageal glands typically overlap the intestine dorsally, and the median bulb is highly developed. Females possess a midline vulva and two equally developed, functional genital tracts, along with rounded spermathecae containing sperm in bisexual species. The tail is elongated, conoid, and approximately 60 µm in length. Males lack a bursa at the tail tip, and their tails are long, conoid, and ventrally arcuate.	

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Table 2 (*continued*)

Genus	Characteristic	Sample images
Genus Hoplolaimus	Members of the genus Hoplolaimus are relatively large nematodes, with females ranging from 1 to 2 mm in length. The labial region is offset, featuring a well-developed cephalic framework that is high, rounded, and heavily sclerotized. The lip region and stylet are also offset, with the stylet bearing anteriorly directed knobs. The esophagus is well-developed, and the lateral fields range from 0 to 4. The vulva is positioned at the mid-body, and the tail is rounded and annulated at the tip. Male Hoplolaimus nematodes are slightly smaller than females. They possess well-developed spicules, a short, arcuate tail, and a bursa that extends to the tip of the tail. The tail is short and bluntly rounded. Scutella, formed by the enlargement of phasmids, are present—one located anterior to the vulva and the other positioned between the anus and vulva [11].	
Genus Hemicycliophora	Genus Hemicycliophora is distinguished by its unique cuticle, which features a detachable sheath, sometimes referred to as a “double” cuticle. The outer layer consists of up to 400 conspicuous annules that are not retrorse. While there is no true lateral field, the cuticle may exhibit various surface ornamentations, including scratches, squares, dots, and longitudinal lines. This genus also shows significant sexual dimorphism. Females are robust, measuring between 0.6 and 1.9 mm in length, with bodies that are either straight or slightly curved ventrally. The anterior end is rounded, while the posterior end is typically pointed, though in rare cases, it may be rounded. The labial region is not distinctly separated from the body and is marked by two to three annules. They possess a long, sturdy stylet with rounded knobs at the base. The procorpus and esophagus contain a well-developed median bulb, while the glands form a small terminal bulb that connects to the intestine. The vulva is located near the posterior end, with the spermatheca positioned laterally and the genital tract oriented anteriorly. The rectum and anus are vestigial. The post-vulval region is usually conical with a pointed end, though in some cases, it may be cylindrical with a rounded tip.	

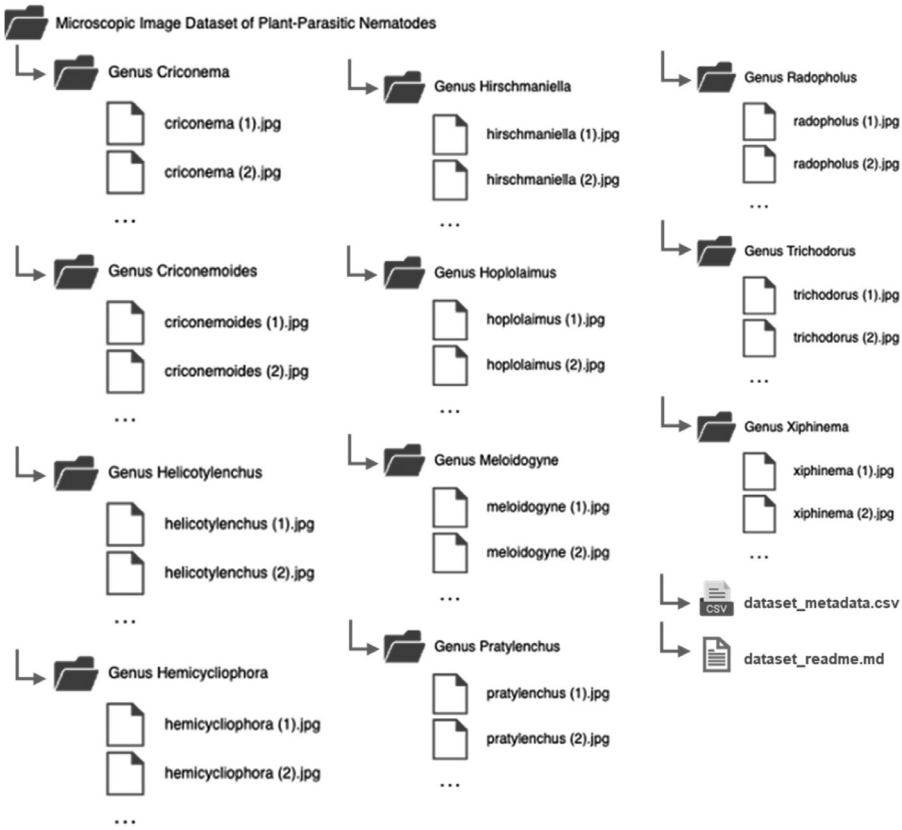

Fig. 2. Organization of the dataset.

equitable use. In the context of agricultural data, issues related to data ownership, informed consent, and benefit sharing are important, especially when the data are derived from smallholder farming communities. In addition, the use of AI in agriculture carries the risk of algorithmic bias, which may result from class imbalance, limited geographic representation, or unaccounted-for contextual variables. Such biases can lead to misclassification or unequal performance in different agricultural environments. Therefore, promoting transparency, fairness, and inclusivity in data collection and AI deployment is critical. Engaging local stakeholders, including farmers and agricultural extension agents, is essential to ensure that AI-driven insights are contextually relevant and ethically grounded.

4. Experimental Design, Materials and Methods

4.1. Location

Data were collected across Central Java and Yogyakarta, Indonesia, between 2020 and 2023. Tomato samples were obtained from Kejajar, Wonosobo, Central Java. Banana and rice samples were collected from Berbah, Sleman, Yogyakarta. Eggplant samples were gathered from two locations: Berbah, Sleman, and Bugel, Panjatan, Kulonprogo, Yogyakarta. Additionally, melon and chili samples were sourced from Ngipiksari, Pakem, Sleman, Yogyakarta. The detailed locations of each sample collection site are listed in Table 3.

Table 3
Sample plant location.

Sample Crop Plant	Location	Coordinate	Collection Date
Tomato	Kejajar, Wonosobo, Central Java	Latitude: -7.2092, Longitude: 109.9244	August 23rd, 2023
Banana and Rice	Berbah, Sleman, Yogyakarta	Latitude: -7.8054, Longitude: 110.4428	August 1st, 2022
Eggplant	Berbah, Sleman, Yogyakarta	Latitude: -7.8054, Longitude: 110.4428	June 20th, 2020
Eggplant	Panjatan, Kulonprogo, Yogyakarta	Latitude: -7.9257, Longitude: 110.1593	June 28th, 2022
Melon and Chilli	Pakem, Sleman, Yogyakarta	Latitude: -7.6065, Longitude: 110.4240	July 15th, 2022

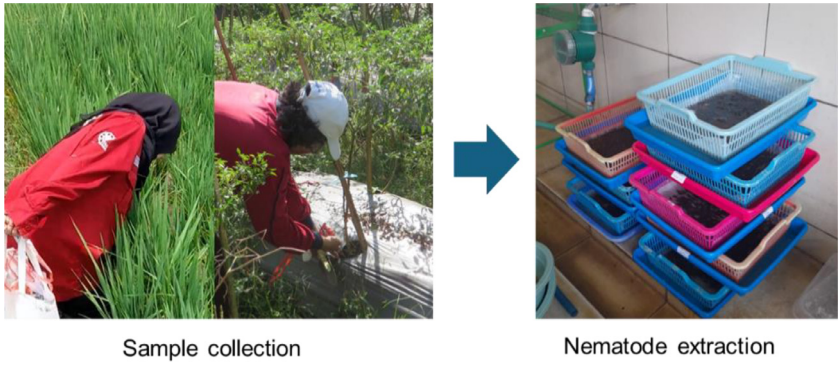

Fig. 3. Sample collection and nematode extraction.

4.2. Sample collection and nematode extraction

Roots from various crops, including rice, tomatoes, melons, chili, and eggplants, exhibiting symptoms of nematode infestation, along with the surrounding soil, were collected from the field to obtain nematode specimens. All samples were obtained through purposive sampling, targeting locations that exhibited suspected or visible symptoms of nematode infestation. The extraction process followed a modified Whitehead-Tray method for the active mobile nematodes [12]. Root samples were finely cut into 1 mm pieces and placed on filter paper supported by a nylon screen inside an inverted bucket, allowing the water to gradually moisten the samples. This setup facilitated the natural migration of nematodes into the water, where they were collected in tubes for further morphological analysis. Extraction was conducted at room temperature (25–30°C) for 24 h to ensure optimal nematode recovery. In addition to the Whitehead-Tray method, female nematodes were extracted directly from root galls using a dissection approach [13]. Root samples were first washed thoroughly and placed in petri dishes containing sterile water. Dissection was performed under a dissecting microscope (15–50 × magnification) using a scalpel, dissecting needles, and fine forceps. Female nematodes embedded in the root tissues were carefully removed and transferred for fixation and identification. Fig. 3 shows the sample collection, along with the extracted nematodes.

Prior to microscopic examination, nematodes were killed and fixed using cooled Formalin Acetic Acid (FAA) to preserve their structural integrity for accurate identification [14]. FAA is a commonly used fixative for nematode preservation, as it helps maintain fine structural details. Proper fixation using chilled FAA helps to maintain nematode morphology by reducing shrinkage. To fix the nematodes using FAA, we prepared a fixative solution consisting of 10 mL formalin (37%), 5 mL glacial acetic acid, 50 mL ethanol (70%), and 35 mL distilled water, which was then cooled to 4°C before use. The extracted nematodes were pre-relaxed by placing them in warm water (55–60°C) for 1–2 min to prevent contraction. Subsequently, they were transferred to a vial containing chilled FAA, ensuring that they were fully immersed in at least ten times their volume of fixative. Small nematodes required 12–24 h for fixation, whereas larger nema-

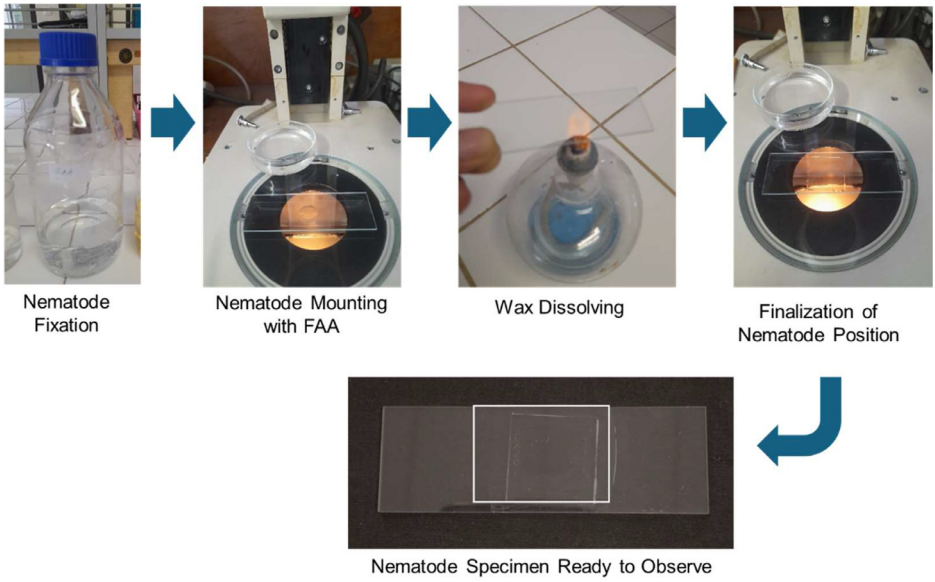

Fig. 4. Preparation nematode specimen for microscopic morphological identification.

todes required up to 48 h at 4°C for optimal preservation. After fixation, the nematodes were washed with 70% ethanol to remove excess FAA and stored in ethanol for long-term preservation. For microscopic observation, the specimens were cleared, stained, and mounted for slide preparation.

Fig. 4 shows the step-by-step process of preparing a nematode specimen slide for microscopic observation. The process began with the preparation of a nematode fixative solution, in which the specimen was preserved in a small dish. In the second step, a paraffin ring was placed on a clean glass slide, followed by the addition of a drop of FAA. The nematode was then carefully hooked and placed in FAA solution. The slides were then heated on a warming plate to enhance adhesion. Simultaneously, a coverslip was briefly passed over a flame using a spirit lamp to remove moisture and prevent the formation of air bubbles during mounting. Once the slide was warmed, the nematode specimen was carefully positioned, and a preheated coverslip was placed over it to ensure proper mounting. The final step, as shown at the bottom of the image, involved the preserved nematode specimen, which was ready for microscopic observation. A comprehensive step-by-step protocol outlining the necessary equipment and parameter settings for nematode sampling, extraction, and slide preparation is available in the Procedural Guidelines provided in Supplementary Material 1.

4.3. Microscopic observation and image capturing

Nematode identification was performed at the Nematology Sub-Laboratory, Plant Pest Science Laboratory, Department of Plant Protection, Faculty of Agriculture, Universitas Gadjah Mada, Yogyakarta, Indonesia. Observations were performed using a light microscope (Olympus CX-31) at magnifications ranging from 4x to 100x to enable a detailed examination of nematode morphology. Images were captured using a digital camera operating in the auto-calibration mode, in which settings such as exposure time were automatically adjusted and not manually recorded. Identification was based on key morphological characteristics, including the position of the mouth, stylet and knob, tail, cuticle, esophagus and intestine, and reproductive organs, following the descriptions provided by Mai and Lyon [15] and Mekete et al. [16].

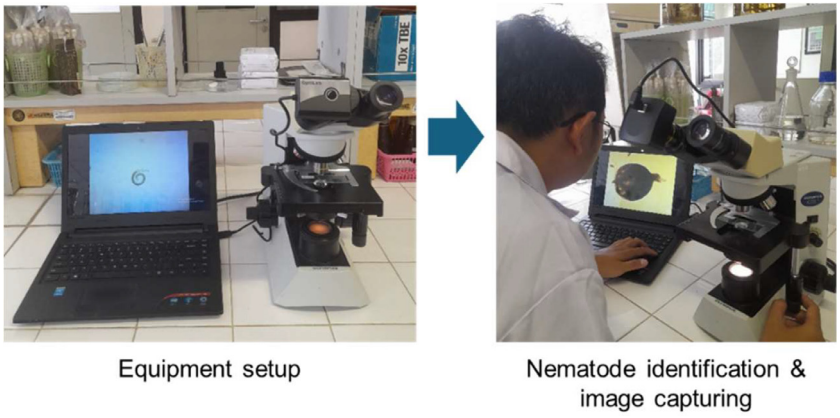

Fig. 5. Equipment set up for nematode identification.

Table 4
Optical cameras specification.

Optical Camera	Specification	Software
Optilab Advance	1/3" color CMOS sensor, Automatic exposure	Optilab Viewer 2.2 & Image Raster 3.0
Optilab Advance Lite	1/2 8" Sony IMX307 sensor, Automatic and manual 0.1ms to 2000ms exposure	Optilab Viewer 4.0 & Image Raster 3.0

Images were captured using an optical system connected to both the microscope and laptop, with the equipment setup illustrated in Fig. 5. The images used in this dataset were captured from multiple perspectives to ensure a comprehensive documentation. Nematode image capture was conducted using two different types of optical cameras, with the specifications and features of each camera provided in Table 4.

The images were then labelled and organized by expert nematologists, with each folder representing a distinct nematode genus. This folder-based labelling approach ensures accurate grouping based on genus-level identification. A detailed step-by-step protocol, including the required equipment and parameter settings for nematode sampling, extraction, slide preparation, and imaging, is provided in the *Procedural Guidelines* in Supplementary Material 1.

4.4. Image processing

The initial captured dataset varied in both image format and resolution. Images are available in JPG and PNG formats with resolutions ranging from 1280 × 1024 pixels to 4100 × 3075 pixels. To standardize the dataset for consistency, all images were converted to JPG format with a uniform resolution of 1280 × 1024 pixels.. The pixel size was selected to preserve sufficient detail for an accurate analysis. Image processing was performed using the Pillow library, a Python-based imaging library that provides tools for image loading, editing, and format conversion [17].

For use in AI applications, particularly in training deep learning models, users may need to further preprocess the images by resizing them to a fixed input resolution (e.g., 224 × 224 pixels) to comply with model architecture requirements, such as those of CNNs. To enhance uniformity and reduce computational complexity, users may also convert images to grayscale. This transformation preserves essential morphological features relevant for nematode identification while simplifying the computational requirement for the training process.

Detailed instructions on how to access and utilize the nematode image dataset, along with a sample Python script are provided in Supplementary Material 2, and can also be accessed directly at: https://github.com/nhs-research/parasitic-nematode-EfficientNetV2B0.

Limitations

This dataset has an imbalanced class distribution, where certain genera contain significantly fewer samples than others. To address this, additional preprocessing techniques, such as data augmentation, are necessary to improve model generalization and classification performance. In addition, the dataset is currently limited to samples collected solely from Indonesia, which may not fully represent the global diversity of plant-parasitic nematodes. Future work will expand the dataset by incorporating samples from different geographic regions to enhance its robustness and make AI-based nematode identification models more widely applicable across diverse agricultural environments.

Ethics Statement

The current work does not involve human subjects, animal experiments, or any data collected from social media platforms.

Data Availability

Microscopic Image Dataset of Plant-Parasitic Nematodes (Original data) (Mendeley Data).

CRediT Author Statement

Siwi Indarti: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation; **Nabila Husna Shabrina:** Conceptualization, Software, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Visualization; **Rina Maharani:** Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative Ai and Ai-assisted Technologies in the Writing Process

During the preparation of this manuscript, the authors used ChatGPT 4o and Paperpal to improve the language and readability of the article. After using these tools, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Supplementary Materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.dib.2025.111687.

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